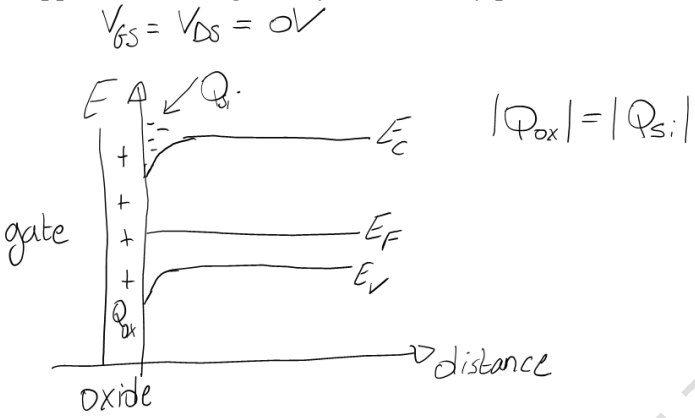
AED exam 2026 answers

1. Compulsory question.

a)

(i) nMOS [1]

(ii) Trapped oxide charge. They are normally positive.



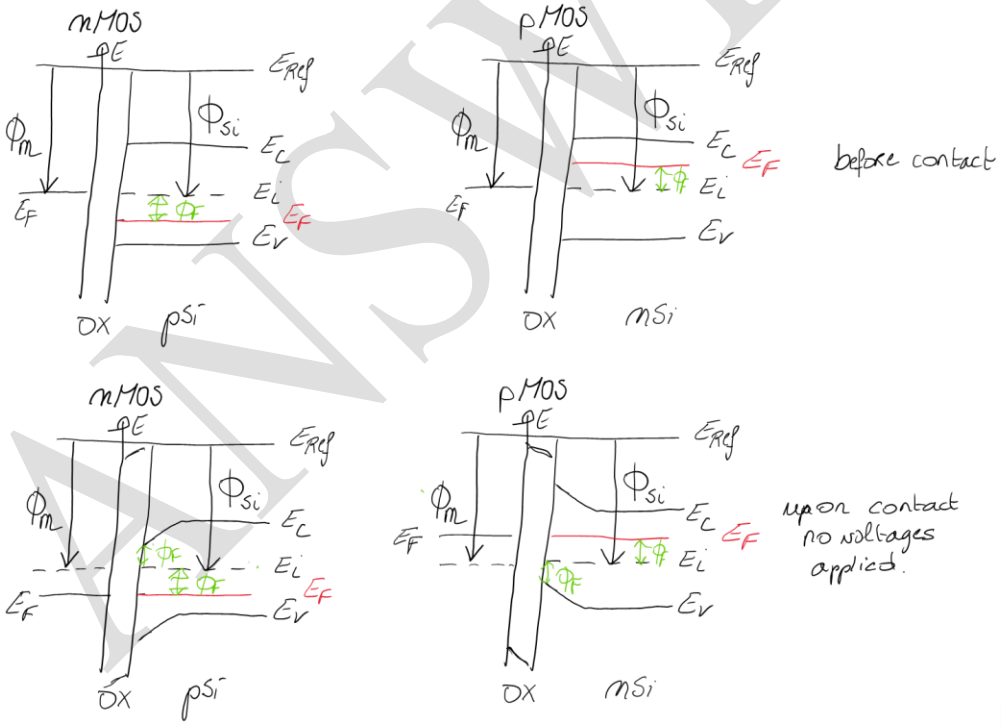
[4]

(iii) Coulomb scattering & surface roughness scattering. [2]

(iv) ionised impurity scattering, the parasitic resistance of the source. [3]

b) .

(i)



[3+3]

(ii) $|V_{thp}| = |V_{thn}|$ they have equal values (nMOS positive and pMOS negative). [2]

(iii) By making a heterojunction of different thin metal layers with specific thickness. This gives a weighted sum for the workfunction (cfr finFETs), e.g. $\phi_m = \frac{t_1\phi_1 + t_2\phi_2 + t_3\phi_3}{t_1 + t_2 + t_3}$ [2]

c)

(i) The rough interface between the Si channel and the amorph SiO2 layer + the net electric field in the channel applying an electrostatic force to the carriers pulling them towards this interface. [2]

ANSWERS

(ii) Line 6 - Enormal [2]

(iii) gm: NWs suffer most from surface roughness scattering decreasing the mobility thus lowering gm
SS: the subthreshold slope is steepest for the NW because of it excellent electrostatic gate control
I_{on}/I_{off}: NSs have the best I_{on} and increasing the width of the sheet can increase I_{on}, not possible with the other implementations. NWs have the best I_{off} because of best electrostatics. Based on the given IV characteristics it is fair to say that in this case the NWs would have the best ratio (slight lowering of Ion (right axix) but large decrease of Ioff (left axis). [2+2+2]

d)

Assume the wave-function Ψ is represented by a travelling wave:

$$\Psi = A\exp i(kx - \omega t) \ , \quad \text{where } k = 2\pi/\lambda$$

For the free particle in a region with potential energy V:
Total energy = kinetic energy + potential energy

$$\Rightarrow \qquad E = \frac{p^2}{2m} + V$$

where p is the momentum. Furthermore, $p = \hbar k$.

[Marks: 2]

Using the wave-function Ψ, we see that:

$$\frac{\partial \Psi}{\partial t} = -i\omega\Psi \quad \text{and} \quad \frac{\partial^2 \Psi}{\partial x^2} = -k^2\Psi$$

$$\Rightarrow -\frac{\hbar^2}{2m}\frac{\partial^2 \Psi}{\partial x^2} + V\Psi = i\hbar\frac{\partial \Psi}{\partial t}$$

where $\Psi\Psi^*$ represents the probability density.

[Marks: 2]

e) (i) An electron of kinetic energy 5 eV:

$$E = 5 \times 1.6 \times 10^{-19} = 8 \times 10^{-19} J$$

$$\therefore p = \sqrt{2mE} = \sqrt{(2 \times 0.911 \times 10^{-30} \times 8 \times 10^{-19})} = 1.21 \times 10^{-24} kg.m/s$$

$$\therefore p = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{1.21 \times 10^{-24}} = 5.5 \times 10^{-10} m = 0.55\ nm$$

[Marks: 3]

(ii) A photon of energy 5 eV

$$E = h\nu = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{8 \times 10^{-19}} = 2.5 \times 10^{-7} m = 250nm$$

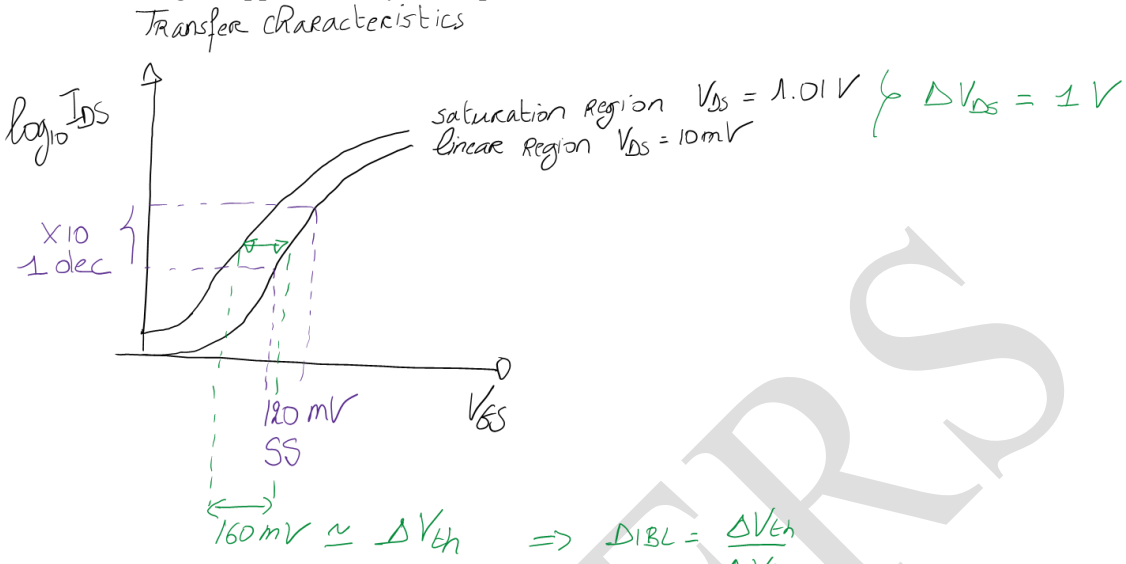
[Marks: 3]

ANSWERS

2. MOSFETs

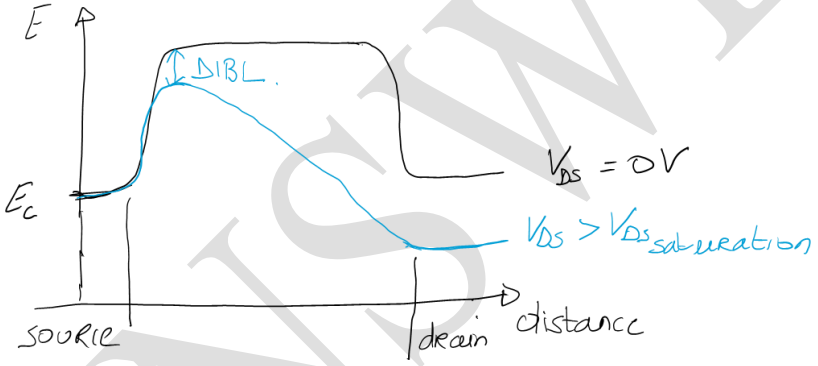
a)

(i) Sketch should give approximately the expected numbers.



[4]

(ii) The potential barrier between the source and channel has reduced due to the electric field from the drain, V_{DS}.

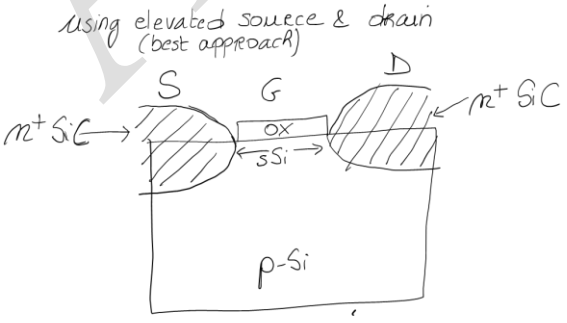


[4]

(iii) Because without that thin thermally grown SiO_2 layer there would be a high level of interface traps that would degrade the performance of the FET. [2]

b)

(i)



[4]

(ii) tensile [2]

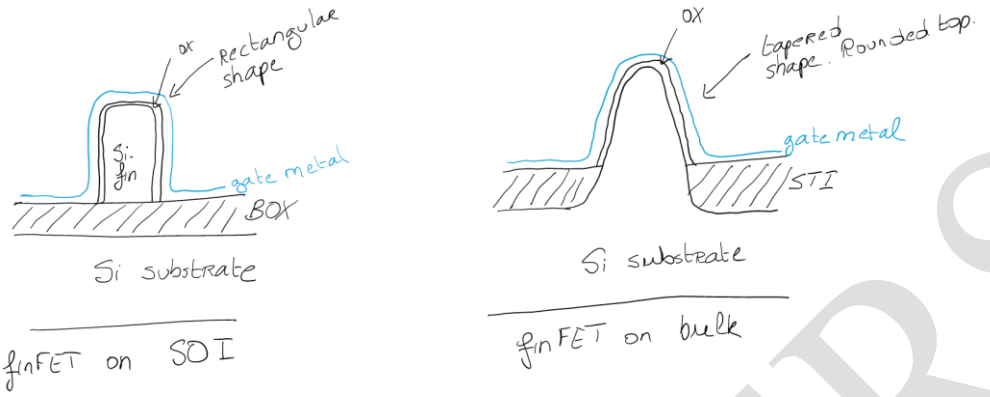
ANSWERS

(iii) The **mobility**. Tensile strain will cause the degenerated conduction bands of Si to split. The 2 bands with the **lowest effective mass** (sharpest parabola) lower in energy and thus e^- will populate these bands. Lower mass means higher mobility. In addition, the splitting of the bands reduces interband scattering processes, thus **increases τ**. Having a **quantum well in the conduction band** under tensile strains is beneficial to keeping the e- in the channel. [4]

ANSWERS

3. finFETs.

a) On SOI: strength = better V_{th} control across the whole fin height, better I_{off} and SS, less threshold voltage roll off. Weakness = sharp corners, give higher electric fields and thus more chance of breakdown, more problems with self-heating, high substrate costs.
On bulk: strength = cheaper fabrication, cheaper substrate, are cooler (heat transport via substrate).
Weakness = poorer carrier concentration control, lower I_{on} and higher I_{off}.



[3+3+2]

b)

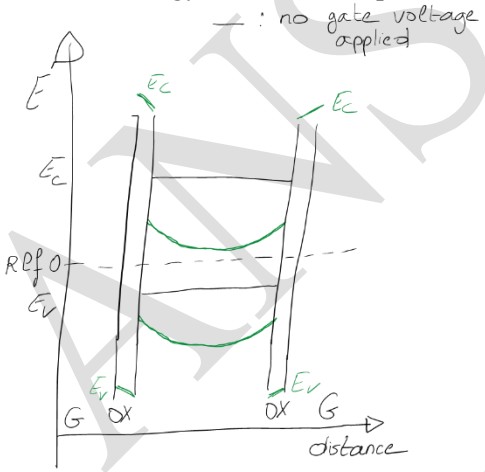
(i) A fin is fully depleted and double gated. FD means there is limited variation in the depletion charge with gate voltage, making $C_{depl} << C_{ox}$. Since 2 gates we actually have $2 \times C_{ox}$ [2]

(ii) By optimising the fabrication process (cleaning processes and environment). [2]

(iii) ~ 59 mV/dec (60 is also acceptable). [2]

c) .

(i) Because the distance between the left and the right gate is very small, their electric fields couple and lower the energy bands with respect to the "zero" reference level. [2]

(ii) 

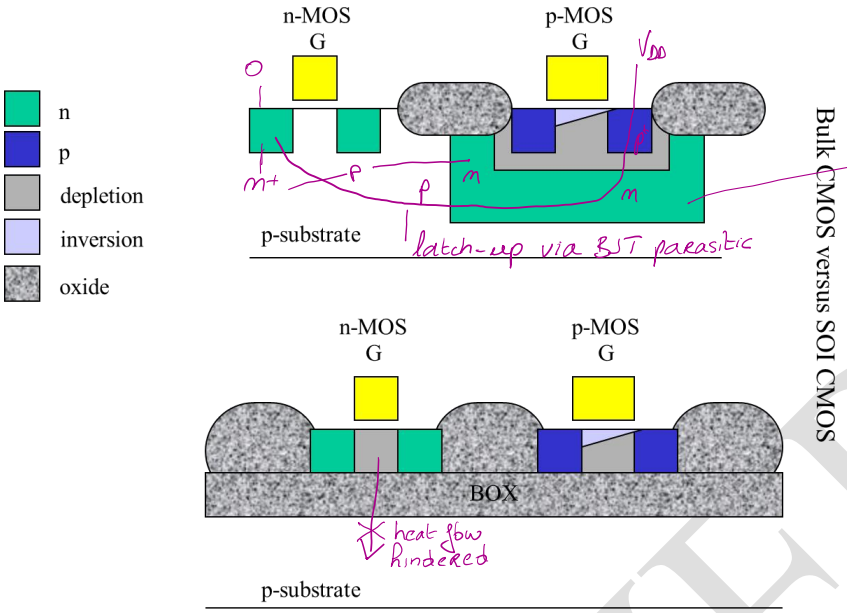
[4]

Note that this electrostatic coupling was explained for FDSOI in the powerpoint slides of that lecture.

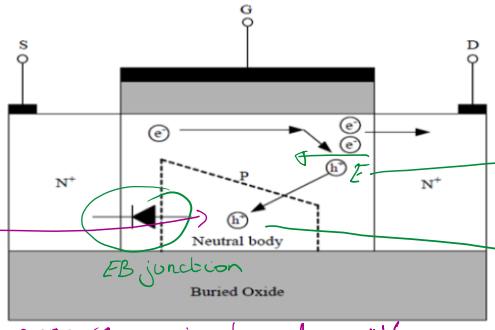
ANSWERS

4. SOI.

a) SOI: advantages simple process -> lower costs, no substrate leakage, elimination of latch up.
Disadvantage: high substrate cost, self-heating [3+3+2]



b) [4]



alternative explanation is that charging the neutral region = similar to a back gate effect. ↑↑ on back gate helps increase -= in channel => ΔV_{th}.

high electric field (V_{DS}) causes impact ionisation e⁻ go to drain but h+ go to neutral body (not under gate control).

holes accumulate here D-neutral-S ≡ BJT npn neutral region ≡ base. Injection of carriers h+ in base cause BE to forward bias => sudden increase in output current.

c) FDSOI

(i) In the formulae list we find the relevant equations [4]

$$V_{th} = \phi_m - \phi_s + 2\times\phi_F + \gamma\times\sqrt{2\times\phi_F}$$

$$\phi_F = \frac{kT}{e}\ln\left(\frac{N_A}{n_i}\right)$$ Fermi energy

$$\gamma = \frac{\sqrt{2e\varepsilon_s N_A}}{C_{ox}}$$ body factor is related to the depletion width and thus charge.

OR Rewritten: $V_{th} = V_{FB} + 2\phi_F + \frac{Q_{depl}(2\phi_F)}{C_{ox}}$

Given is that we work with FDSOI, thus the depletion charge variation as a function of gate voltage will be very small. Also given is that the BOX layer is thick, thus the coupling between the grounded substrate and the channel will be negligible. Based on these assumptions, the body effect can be neglected, thus simplifying the expression of V_{th} to: $V_{th} = V_{FB} + 2\phi_F$

(ii) In UTBB-SOI there is a very strong coupling between the substrate and the drain across the large drain-substrate capacitance. This coupling will influence the value of the threshold voltage and the expression for V_{th} becomes much more complex. [2]

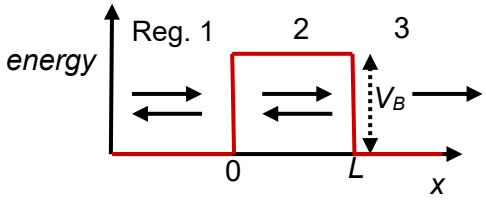
d) Smaller: $V_{thD} < V_{th0}$ [2]

ANSWERS

5. Quantum devices

(a) Here, the potential energy $V(x)$ given by:

$V(x) = V_B,\ 0 \le x \le L$
$V(x) = 0,\ x < 0,\ x > L$



The general, space-dependent wave functions in regions 1, 2 and 3 are:

Region 1:

$$\psi_1 = a_{1i}\exp(ik_1x) + a_{1r}\exp(-ik_1x) = \exp(ik_1x) + r\exp(-ik_1x)$$

where we assume $a_{1i} = 1$
and r = amplitude reflection coeff.

Region 2:

$$\psi_2 = a_{2i}\exp(ik_2x) + a_{2r}\exp(-ik_2x)$$

Region 3:

$$\psi_3 = a_{3i}\exp(ik_1x)$$

Give 1 marks per wave function.

[Marks: 3]

(b) Wave vectors in regions 1 and 2, for energy E, are as follows:

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}} \qquad k_2 = \sqrt{\frac{2m(E-V_0)}{\hbar^2}}$$

As region 3 is identical to region 1, therefore $k_3 = k_1$

Give 2 marks per wave vector.

[Marks: 6]

(c) The boundary conditions at $x = 0$ are:

$$\psi_1 = \psi_2 \quad \text{and} \quad \frac{d\psi_1}{dx} = \frac{d\psi_2}{dx}$$

The boundary conditions at $x = L$ are:

$$\psi_2 = \psi_3 \quad \text{and} \quad \frac{d\psi_2}{dx} = \frac{d\psi_3}{dx}$$

Applying these to the general wave functions in (a) then gives us:

At $x = 0$:

$$\psi_1 = \psi_2 \quad \Rightarrow \quad 1 + r = a_{2i} + a_{2r} \qquad (1)$$

$$\frac{d\psi_1}{dx} = \frac{d\psi_2}{dx} \quad \Rightarrow \quad k_1(1-r) = k_2(a_{2i} - a_{2r}) \qquad (2)$$

[Marks: 2]

ANSWERS

Similarly, at $x = L$:

$$a_{2i}e^{ik_2L} + a_{2r}e^{-ik_2L} = a_{3i}e^{ik_1L} = t \qquad (3)$$

$$k_2\left(a_{2i}e^{ik_2L} - a_{2r}e^{-ik_2L}\right) = k_1 t \qquad (4)$$

[Marks: 3]

(d) Dividing Eq. 4 by k_2 and then adding to Eq. 3 gives:

$$a_{2i} = \left(1 + \frac{k_1}{k_2}\right)\frac{te^{-ik_2L}}{2} \qquad (5)$$

Substituting Eq. 5 into Eq. 3 gives:

$$a_{2r} = \left(1 - \frac{k_1}{k_2}\right)\frac{te^{ik_2L}}{2} \qquad (6)$$

[Marks: 3]

Dividing Eq. 2 by k_1, and then adding to Eq. 1 gives:

$$2 = a_{2i}\left(1 + \frac{k_2}{k_1}\right) + a_{2r}\left(1 - \frac{k_2}{k_1}\right) \qquad (7)$$

Finally, substituting Eq. 5 and 6 into Eq. 7 and simplifying gives the amplitude transmission coefficient t:

$$\boxed{t = \frac{4k_1k_2}{(k_1 + k_2)^2 e^{-ik_2L} - (k_1 - k_2)^2 e^{ik_2L}}} \qquad (8)$$

[Marks: 3]

ANSWERS